

15/01/25

Lecture 3

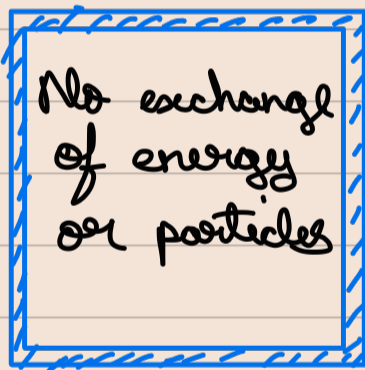
Aim:

- Ensembles & Equivalences
- Quantum Stat. (Fermi & Spins)
- Response functions & their relation with fluctuations

Hydrodynamic System

$$E = \varepsilon V$$
$$N = \rho V$$

Microcanonical



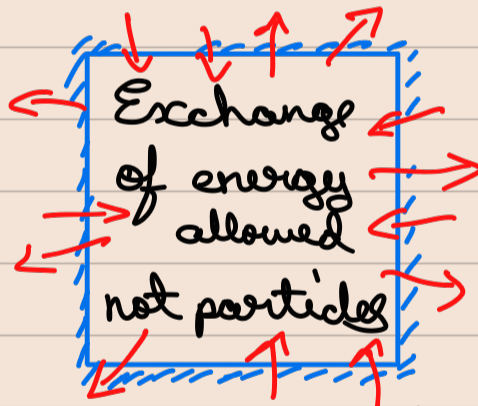
$$(N, V, E)$$

' Ω ' is found

$$S = k_B \ln(\Omega)$$

↳ Entropy

Canonical



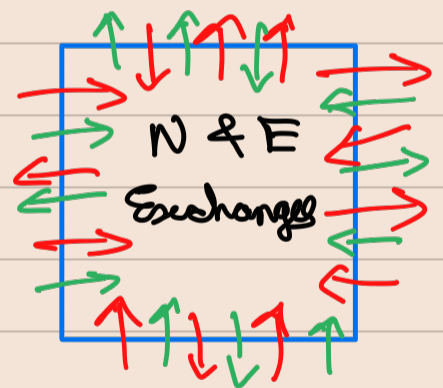
$$(N, V, T)$$

' Q ' is found

$$A = -k_B T \ln(Q)$$

↳ Helmholtz
free energy

Grand Canonical



$$(\mu, V, T)$$

' Z ' is found

$$\Lambda = -k_B T \ln(Z)$$

$\Lambda = -PV$
↳ Grand potential

Equivalence of Ensembles

(For Hydrodynamic System)

Relating Canonical with grand canonical

For Grand Canonical Ensemble,

$$e^{-\beta\Omega} = \mathcal{Z} = \sum_N e^{\beta\mu N} Q(N, V, T)$$

But $Q = e^{-\beta A_c}$ ($A_c \Rightarrow$ Canonical ensemble Helmholtz free energy)

$$\Rightarrow \mathcal{Z} = \sum_N e^{\beta(\mu N - A_c)} \quad N = PV$$

$$\Rightarrow \mathcal{Z} = \sum_N e^{-\beta V(a_c - \mu P)} \quad a_c = \frac{A_c}{V}$$

In thermodynamic limit ' $V \rightarrow \infty$ '
So only that ' P ' for which the
function $\mu P - a_c$ is maxima will
dominate.

$\Rightarrow$ Maxima of $\mu P - a_c$ w.r.t ' P '

$$\text{is } \left. \frac{\partial}{\partial P} (\mu P - a_c) \right|_{P=\bar{P}} = 0$$

$$\Rightarrow \left(\frac{\partial a_c}{\partial p} \right)_{p=p_c} = \mu_c$$

Thus at thermodynamic limit as $V \rightarrow \infty$, in the summation

$$Z = \sum_n e^{-\beta V(a_c - \mu p)}$$

that term where $(\mu p - a_c)$ is maxima will dominate. Thus,

$$Z \sim e^{-\beta V(a_c - \mu_c p_c)} = e^{-\beta(A_c - \mu_c N_c)}$$

$$\frac{\ln(Z)}{\beta} = -A_c + \mu_c N_c$$

Now $A_c = E_c - T_c S_c$ $E_c = T_c S_c - P_c V_c + \mu_c N_c$

$$\Rightarrow A_c = -P_c V_c + \mu_c N_c$$

$$\Rightarrow \underline{-A_c + \mu_c N_c = P_c V_c}$$

But $P_{g.c.} V_{g.c.} = k_B T \ln(Z) = \frac{\ln(Z)}{\beta}$

$$\Rightarrow \underline{P_{g.c.} V_{g.c.} = P_c V_c} \Rightarrow \text{only at Thermodynamic limit}$$

When Quantum Stat. is Important?

“When the de broglie wave-length of the particles is more than the average inter particle spacing, then interference effects starts to dominate & then we should use quantum statistics.”

$$\lambda_d = \frac{h}{p}$$

$$\frac{p^2}{2m} = K_B T$$

$$p = \sqrt{2mK_B T}$$

$$\Rightarrow \lambda_d = \frac{h}{\sqrt{2mK_B T}}$$

$$\lambda_{Th} = \frac{h}{\sqrt{2m\pi K_B T}}$$

(Thermal.
de. broglie
wavelength)

Let $l \Rightarrow$ Average Interparticle spacing

for $\lambda_{Th} \geq l \Rightarrow$ Q.M. stat.

$\lambda_{Th} < l \Rightarrow$ Classical stat.

Eg:- For Helium, at $T = 2\text{ K}$
 $m_{He} \sim 4u$

Compare λ_{Th} & l

$$u = 1.66 \times 10^{-27} \text{ Kg.}$$

$$\Rightarrow \lambda_{Th} = 6.172 \text{ \AA}$$

So interparticle distance

b/w Helium particles has to be

less than $6.172 \text{ \AA}$, Since Helium

Shows quantum properties of

Bose-einstein Condensation.

Operators & Eigen-kets

Formalism of Quantum mechanics relies on Linear algebra. The state of the quantum system is denoted by a ket that lives in an abstract Hilbert space. Every observable is an operator which has eigen values & eigen kets.

Hamiltonian operator $\Rightarrow \hat{H}$

Energy E_n are eigen values of $\hat{H}$.

$$\Rightarrow \hat{H}|n\rangle = E_n|n\rangle$$

$|n\rangle \Rightarrow$ eigen ket.

Thermodynamic probability of Microcanonical ensemble is given

$$\text{by } \Omega(E) = \sum_n \delta(E - E_n).$$

In terms of $\hat{H}$ & eigen kets $|n\rangle$
we can write,

$$\Omega = \sum_n \delta(E_n - E) = \text{Trace}(\delta(\hat{H} - E))$$

~~Proof~~

$$\delta(E_n - E) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{-ik(E_n - E)}$$

now $\hat{H} = \sum_n E_n |n\rangle \langle n|$

$$\Rightarrow f(\hat{H}) |n\rangle = f(E_n) |n\rangle$$

where 'f' is some smooth function that
can be written as a power series.

$$\Rightarrow \langle n | e^{ik(\hat{H} - E)} | n \rangle = e^{ik(E_n - E)}$$

$$\Rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \langle n | e^{ik(\hat{H} - E)} | n \rangle = \delta(E_n - E)$$

$$\Rightarrow \langle n | \delta(\hat{H} - E) | n \rangle = \delta(E_n - E)$$

$$\Rightarrow \Omega(E) = \sum_n \delta(E_n - E) = \sum_n \underbrace{\langle n | \delta(\hat{H} - E) | n \rangle}_{\text{trace}}$$

$$\Rightarrow \boxed{\Omega(E) = \text{trace} [\delta(\hat{H} - E)]}$$

Similarly we can write

$$Q = \sum_n e^{-\beta E_n} = \text{Trace}(e^{-\beta \hat{H}})$$

$$Z_{g.c} = \sum_{N,n} e^{-\beta(E_n - \mu N)} = \text{Trace}[e^{-\beta(\hat{H} - \mu N)}]$$

$$\underset{\substack{\text{(gibbs)} \\ \text{ensemble}}}{Z} = \sum_{n,M} e^{-\beta(E_n - hM)} = \text{Trace}[e^{-\beta(\hat{H} - hM)}]$$

Classical limit for Spin Systems

The quantum nature of certain systems is mainly due to the non-commutation of Spin operators given by $[\hat{S}_\alpha, \hat{S}_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} \hat{S}_\gamma$

Since $\hat{S}^2$ commutes with all the Spin components, whereas none of the Spin components commute, we can completely determine the eigen values of $\hat{S}^2$ & one of Spin component.

Conventionally, $\hat{S}^2$ & S_z are chosen to represent the Spin state.

The unknown quantities are $\hat{S}_x$ & $\hat{S}_y$. Then the deviation is given by, $\hat{S}_x^2 + \hat{S}_y^2 \Rightarrow$ unknown
 $\hat{S}^2$ & $\hat{S}_z \Rightarrow$ known.

$$\Rightarrow \langle \hat{S}_x^2 + \hat{S}_y^2 \rangle = \langle \hat{S}^2 - \hat{S}_z^2 \rangle$$

for a state $|s, m_s\rangle$

$$\langle \hat{S}_x^2 + \hat{S}_y^2 \rangle = (\hbar^2 s(s+1) - m_s^2 \hbar^2)$$

$$\Rightarrow \langle \hat{S}_x^2 + \hat{S}_y^2 \rangle = \hbar^2 (s^2 + s - m_s^2)$$

$$\text{now } \frac{\langle \hat{S}_x^2 + \hat{S}_y^2 \rangle}{\langle \hat{S}^2 \rangle} = \frac{s(s+1) - m_s^2}{s(s+1)}$$

The most minimum value of $\langle \hat{S}_x^2 + \hat{S}_y^2 \rangle$ is when $|\langle \hat{S}_z \rangle|$ max.
i.e. when $m_s = \pm s$

Thus for max. $\langle \hat{S}_z \rangle$ state,

$$\frac{\langle \hat{S}_x^2 + \hat{S}_y^2 \rangle}{\langle \hat{S}^2 \rangle} = \frac{S}{S(S+1)} = \frac{1}{1+S}$$

$$= \frac{1}{S} \left[\frac{1}{1+(1/S)} \right]$$

$$= \frac{1}{S} \left[1 + \frac{1}{S} + \frac{1}{S^2} + \dots \right]$$

$$= \frac{1}{S} + \frac{1}{S^2} + \dots$$

The relative root mean square deviation or the angular uncertainty is given by $\sqrt{\frac{\langle \hat{S}_x^2 + \hat{S}_y^2 \rangle}{\langle \hat{S}^2 \rangle}} = \frac{1}{\sqrt{S}} \left(1 + \frac{1}{2S} + \dots \right)$

Thus in the limit of $S \rightarrow \infty$ we may consider spins to have definite direction.

Gibb's Canonical ensemble of Spin Systems

Consider a System of N -Spins. The interacting Spins are represented by

$$\hat{H} = \sum_{\substack{i,j \\ i \neq j}} J_{ij} \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j$$

The gibb's Canonical partition function is given by,

$$Z \rightarrow Z(\beta, h, N) = \text{Trace} \left[e^{-\beta(\hat{H} - hM)} \right]$$

assuming $[\hat{H}, M] = 0$

Average magnetization is given by,

$$\langle M \rangle = \frac{\text{Trace} [M e^{-\beta(\hat{H} - hM)}]}{Z}$$

$$\langle M \rangle = \frac{1}{\beta Z} \left[\frac{\partial}{\partial h} \left(\underbrace{\text{Trace} [e^{-\beta(\hat{H} - hM)}]}_Z \right) \right]_{\beta, N}$$

$h \Rightarrow$ represents the ext. magnetic field needed to achieve given magnetization.

$$\Rightarrow \boxed{\langle M \rangle = \frac{1}{\beta} \left[\frac{\partial}{\partial h} \ln(Z) \right]_{\beta, N}}$$

Magnetic Susceptibility (χ_m)

It is defined as $\chi_m = \left(\frac{\partial M}{\partial h} \right)_{\beta, N}$

$$\Rightarrow \chi_m = \frac{\partial}{\partial h} \left[\frac{1}{\beta} \frac{\partial}{\partial h} \ln(Z) \right]_{\beta, N}$$

$$\boxed{\chi_m = \frac{1}{\beta} \left[\frac{\partial^2}{\partial h^2} \ln(Z) \right]_{\beta, N}}$$

$$\Rightarrow \chi_m = \frac{1}{\beta} \frac{\partial}{\partial h} \left(\frac{1}{Z} \left(\frac{\partial Z}{\partial h} \right)_{\beta, N} \right)_{\beta, N}$$

$$\chi_m = \frac{1}{\beta} \left[\frac{1}{Z} \left(\frac{\partial^2 Z}{\partial h^2} \right)_{\beta, N} - \frac{1}{Z^2} \left(\frac{\partial Z}{\partial h} \right)_{\beta, N}^2 \right]$$

$$= \frac{1}{\beta} \left[\frac{1}{Z} \text{Trace} \left(\beta^2 M^2 e^{-\beta(H - hM)} \right) - \beta^2 \langle M \rangle^2 \right]$$

$$\Rightarrow \chi_m = \beta \left[\langle M^2 \rangle - \langle M \rangle^2 \right] = \beta \langle (\Delta M)^2 \rangle$$

$$\text{where } \underline{\Delta M = M - \langle M \rangle}$$

Magnetic Susceptibility per Spin is defined as

$$\chi_s = \left(\frac{\partial m}{\partial h} \right)_{\beta, N}, \quad m = \frac{M}{N}$$

$$\Rightarrow NX_s = \left(\frac{\partial M}{\partial h} \right)_{B, N}$$

$$\& \underbrace{\langle (\Delta M)^2 \rangle}_{\text{variance in } M} = k_B T NX_s$$

$$\sigma_m \Rightarrow \text{std. deviation of } M = \sqrt{\langle (\Delta M)^2 \rangle}$$

$$\boxed{\sigma_m = \sqrt{N k_B T X_s}}$$

Relative RMS fluctuation in M is $\frac{\sigma_m}{\langle M \rangle}$

$$\langle M \rangle = N_\uparrow - N_\downarrow \propto N$$

$N_\uparrow \Rightarrow$ no. of up spins
 $N_\downarrow \Rightarrow$ no. of down spins

$$\Rightarrow \text{Fluctuation} \propto \frac{\sqrt{N T X_s}}{N} \propto \sqrt{\frac{T X_s}{N}}$$

For finite X_s

When $N \rightarrow \infty$, Relative RMS fluctuation $\rightarrow 0$

For $X_s \rightarrow \infty$

Here both N & $X_s \rightarrow \infty$. So let

$$X_s \propto N^\phi \Rightarrow \text{Fluctuation} \propto \frac{1}{N^{(1-\phi)/2}}$$

$\phi \Rightarrow$ some exponent.